

Bounded resource profile (not benchmark data)

Link parser

$O(n)$, fixed 531-byte frame bound

Address allocator

at most 254 probes

Route lookup

small fixed table, linear scan

work

No dynamic growth

frame pool + static slots

stream parse: linear

fixed-capacity tables: bounded

input / configured capacity

Actual latency depends on UART topology and kernel workload; no fabricated throughput values are shown.